

FGI-GSRx v2.1.3 – Release Notes

1. New Feature in v2.1.3

1.1 GPS L1 C/A signal tracking with 20-milliseconds (ms) integration

The user needs to choose the receiver tracking mode for GPS L1 by setting the following parameter in the receiver configuration file (for example, 'defaultReceiverConfiguration.txt' or user's own defined receiver configuration file):

%Choose tracking mode: '1msTracking' or '20msTracking'

gpsl1, trackingMode, '20msTracking',

A new '20msTracking' mode is introduced for GPS L1 C/A signal to offer 20-ms integration based tracking after bit synchronization. The receiver first estimates the bit boundaries and then switches to fine tracking starting from the first detected bit interval that allows 20 ms integration without having any bit transition during the correlation interval.

If the user chooses '20msTracking' mode, all the required FLL, PLL and DLL tracking parameters will be set considering 20 ms coherent integration at the tracking stage. The FLL, PLL and DLL will be invoked after 20 ms instead of 1 ms, and therefore, all the tracking loop parameters are adjusted accordingly in a newly introduced function named as:

'gpsl1UpdateTrackingParametersFor20msIntegration.m'

The user can change tracking loop parameters within this function, if need be, considering signal correlation after every 20 ms.

(1)Changes: v2.1.2 => v2.1.3

Folder\Function Name	Remarks	Modification Type
\param\gpsl1UpdateTrackingParametersFor20msIntegration.m	Updated tracking parameters for 20ms tracking. This function is only called after the bit synchronization when it switches from '1msTracking' to '20msTracking' mode.	New function added
All the shared parameter *.txt files are updated with the new changes. \param\default_param_GPSL1_Chapter2.txt \param\default_param_MultiGNSS_Chapter7.txt	=> A new '20msTracking' mode is introduced for GPS L1 C/A signal to offer 20-ms integration based tracking after bit synchronization. => One new DLL tracking parameter is introduced: 'dllLoopGain'. => FGI-GSRx version number is included to keep track of the receiver's version number.	

\\main\\gsrx.m	The function is modified to accommodate '20msTracking' after it finds bit boundaries with '1msTracking'.	Modified
\\track\\doBitSynchronization.m	It does bit synchronization within the first few seconds of data for GPS L1 C/A signal.	New function added
\\track\\bit\\bitSync.m	The receiver attempts for bit synchronization before 1.5 seconds in order to allow the receiver to settle down from pull-in to coarse-tracking mode. Instead of looking for a bit depth of 2 bits for bit synchronization, it now looks for 5 bits for the bit synchronization: which makes it take a better decision on the bit boundary index than before, especially in relatively weak signal condition.	
\\track\\allocateTrackChannel.m	A new tracking variable is introduced to estimate Signal-to-Noise Ratio (SNR). In addition, 'dllLoopGain' is now taken as receiver input configuration parameter.	Modified
\\track\\allocateTrackChannelHeader.m	Tracking state is set to 'STATE_FINE_TRACKING' already from the beginning for GPS L1 C/A '20msTracking' mode.	Modified
\\track\\gpsl1UpdateChannelState.m	Changes on the tracking channel update states: it is solely based on the estimated FLL and PLL lock indicator values and the thresholds chosen for those lock indicators.	Modified
\\track\\gpsl1setTrackingTable.m	Accommodates both 1-ms and 20-ms tracking modes. In 20-ms tracking mode, it utilizes	Modified

	'freqDiscrimAtan' instead of 'freqDiscrimAtan2'.	
\track\power\lockDetect.m	Updates due to introduction of '20msTracking' mode.	Modified
\track\power\CN0fromSNR.m	The SNR estimation is now added. Time window for smoothing C/N0 and SNR estimation is fixed to 5000 ms. It can be changed, if needed.	Modified
\track\pll\phaseLoopFilterVeryNarrow.m	Bug is fixed: 'signalSettings' was mistakenly written.	Modified
\frame\gpsl1DecodeEphemeris.m	GPS L1 decode ephemeris function is updated to accommodate both 1-ms and 20-ms tracking modes.	Modified
\frame\doFrameDecoding.m	Changes needed to accommodate '20msTracking' mode.	Modified
\frame\findPreambles.m	Changes needed to accommodate '20msTracking' mode.	Modified
\mod\gpsl1ModulatePrnCode.m	Modifications were done to accommodate '20msTracking' mode.	Modified
\obs\generateObservations.m \obs\getTransmitTime.m \obs\checkObservations.m	- Signal-to-Noise Ratio is now estimated from C/N0 and the coherent integration time used for signal correlation: it is now used as SNR masking. Change from the previous versions (up to v2.1.2), C/N ₀ was used as SNR mask. Now, in this new version, we estimated SNR and used it as an SNR masking criteria.	Modified
\plot\plotTracking.m	Signal-to-Noise Ratio (SNR) plot is also included in the function.	Modified
\stats\calcStatistics.m	=> A small fix on the index (related to smoothing based on 1 Hz update rate). => Number of satellites is also displayed in the ENU plot.	Modified

